

Chapter 6: Principal Component Analysis

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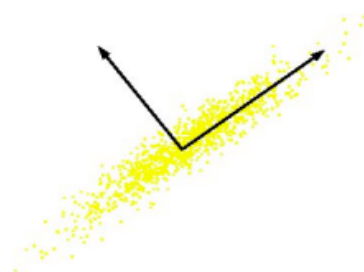
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Overview

- PCA (Principal Component Analysis) is a linear dimension reduction technique



- The idea is to produce a set of variables $Y_1, Y_2, \dots, Y_q$ that are linear combinations of original variables and are uncorrelated, but which are most informative about the data. The thinking is that a variable having a larger variance is more informative/important.

Idea

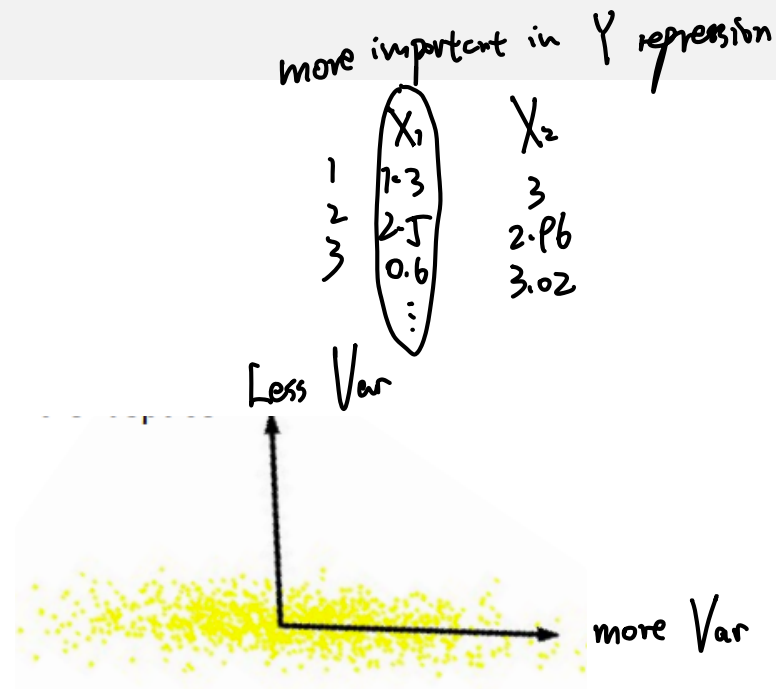


Figure: Intrinsically lower dimensional than the dimension of the ambient space.

$$\begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix} \Rightarrow \begin{aligned} Y_1 &= \sum_{i=1}^p c_{1i} X_i \\ Y_2 &= \sum_{i=1}^p c_{2i} X_i \end{aligned}$$

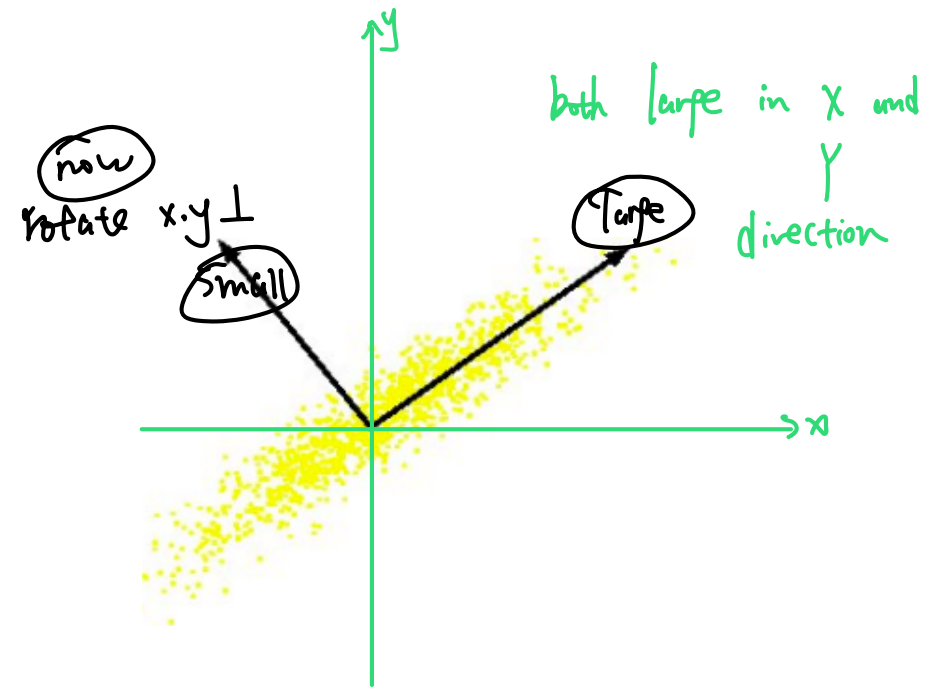
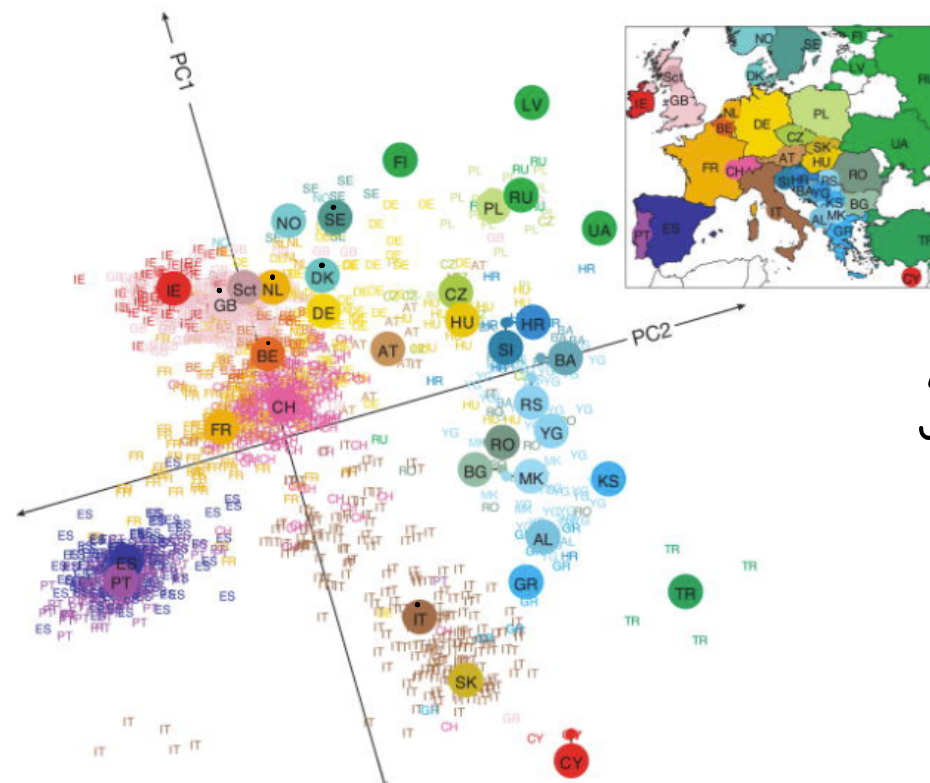


Figure: If we rotate data, again only one coordinate is more important.

Applications

Applications of the PCA method (and many other dimensionality reduction methods)

- More Important {
- **Visualisation:** Make high-dimensional data visible to human eyes
 - **Exploration:** Investigate and discover patterns, structures, and relationships within data
 - **Compression:** Efficient data storage and computation



$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix}$$
 gene for each person-long chain

$$\downarrow \text{PCA}$$

$$Y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Figure: Source: John Novembre, et al. Genes mirror geography within Europe. Nature 456 (2008): 98-101. © 2008.

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Assume $\text{COV}(X) = \Sigma$ and common mean
 $X_1, X_2, \dots, X_n$

3 Sample Principal Components

4 PCs obtained from standardized variables

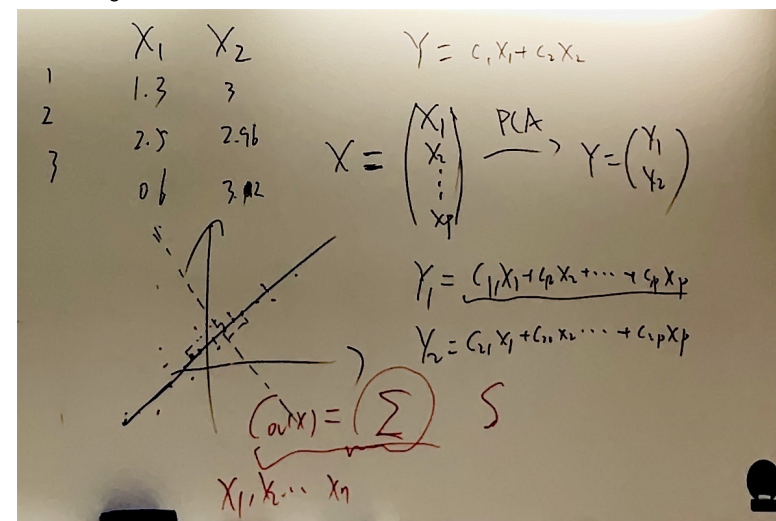
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Population principal components

- All the observations (measurements) are made on the members of the “population”.
 - European countries in one study could be considered the population and you have data for each of them (the variables are percents of people employed in different industries).
 - The psychological test data consist of measurements on 64 subjects. These subjects are a sample from some populations.
- In Population principal components, we can compute Σ and the principal components (PCs) are derived from Σ .

What are PCs

- Algebraically: PCs are linear combination of original variables such that $Y_1 = \mathbf{a}_1^\top \mathbf{X}$ has maximum variance. We then choose $\mathbf{a}_2$ so that $Y_2 = \mathbf{a}_2^\top \mathbf{X}$ has maximum variance subject to being uncorrelated with y_1 , and so on.
- Geometrically: Rotation to new coordinate system.



Geometry of PC

- PCs represent a selection of a new coordinate system obtained by rotating the original axes to a set of new axes (to provide a simpler structure).
- The first principal component represents the direction of maximum variability.
- The second principal component represents the direction of maximum variability that is orthogonal to the first.
- And so on, until the last PC which represents the direction of minimum variability & orthogonal to all of the others.

Axis rotation

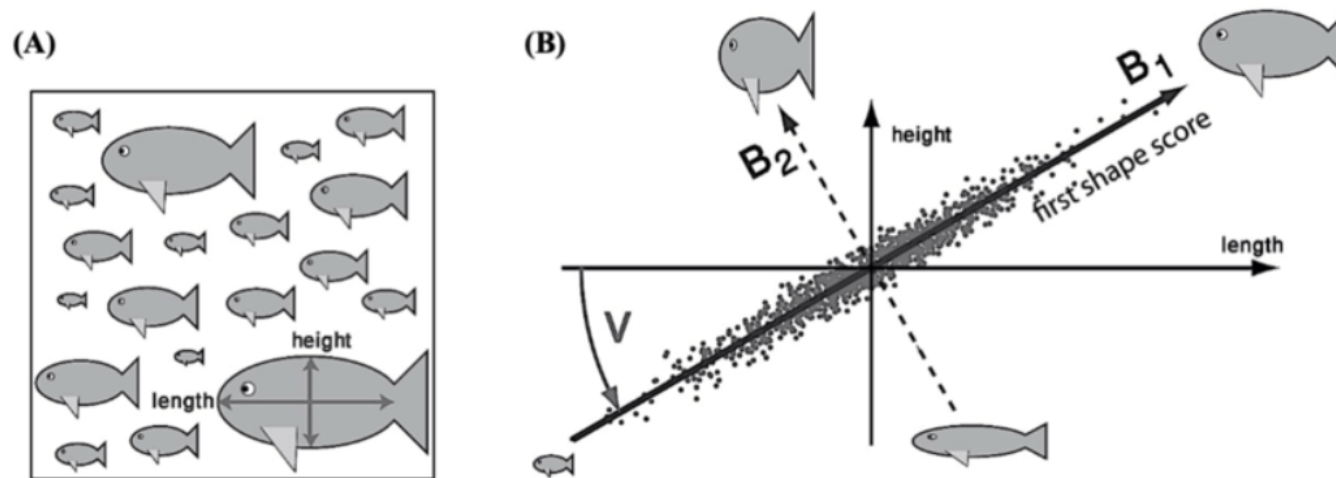


Figure: Height and length of the fishes are assumed to be strongly correlated. PCA defines a change of coordinate system from the original (height, length)-axes (here, the x-and y-axes) to a new axes (B1 and B2) which depict the principle axes of the dimensions that covary. The process of defining a new coordinate system (V) corresponds to a reduction of the dimensionality of the data space, which also retains most of the data's variability.

The algebra of population PCA

We want to transform p variables to q orthogonal linear combinations (generally) where $q \ll p$.

$$\mathbf{X}_{1 \times p}^\top = (X_1, X_2, \dots, X_p) \quad \text{to} \quad \mathbf{Y}_{1 \times q}^\top = (Y_1, Y_2, \dots, Y_q)$$

There are p possible ones

$$Y_1 = \mathbf{a}_1^\top \mathbf{X} = a_{11}X_1 + a_{12}X_2 + \dots + a_{1p}X_p$$

$$Y_2 = \mathbf{a}_2^\top \mathbf{X} = a_{21}X_1 + a_{22}X_2 + \dots + a_{2p}X_p$$

$$\vdots$$

$$Y_p = \mathbf{a}_p^\top \mathbf{X} = a_{p1}X_1 + a_{p2}X_2 + \dots + a_{pp}X_p$$

Given the covariance matrix Σ_X of $\mathbf{X}$, we know

$$\text{var}(Y_i) = \mathbf{a}_i^\top \Sigma_X \mathbf{a}_i \quad \text{and} \quad \text{cov}(Y_i, Y_k) = \mathbf{a}_i^\top \Sigma_X \mathbf{a}_k$$

More formal definition of PC

PCs are the uncorrelated linear combinations, $\text{cov}(Y_i, Y_k) = 0$ for all $i \neq k$, with variances as large as possible.

In particular,

$\text{var}(Y_1)$ is the maximum $\rightarrow$ find $\mathbf{a}_1$ that maximizes $\mathbf{a}_1^\top \Sigma_X \mathbf{a}_1$;

$\text{var}(Y_2)$ is the maximum and $\perp Y_1 \rightarrow$ find $\mathbf{a}_2$ that maximizes $\mathbf{a}_2^\top \Sigma_X \mathbf{a}_2$
and $\mathbf{a}_1^\top \Sigma_X \mathbf{a}_2 = 0$

- At each step, select $\mathbf{a}_i$ such that $\mathbf{a}_i^\top \mathbf{X}$ has maximum variance subject to being uncorrelated with all other linear combinations.
- Usually (but not always), we only use $Y_1, Y_2, \dots, Y_q$ where q is much less than p (primary goal is data reduction).

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix} \rightarrow p\text{-dim random vector}$$

Covariance: Σ_X

$$Y_1 = a_1^T X = a_{11}x_1 + a_{12}x_2 + \dots + a_{1p}x_p?$$

$$a_1 = \arg \max_a \text{Var}(a^T X)$$

$$= \arg \max_a a^T \Sigma_X a$$

$$Y_2 = a_2^T X = a_{21}x_1 + a_{22}x_2 + \dots + a_{2p}x_p?$$

$$a_2 = \arg \max_{a, \text{Cov}(Y_1, Y_2) = 0} \text{Var}(a^T X)$$

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix} \rightarrow p\text{-dim random vector}$$

Covariance: Σ_X

$$Y_1 = a_1^T X = a_{11}x_1 + a_{12}x_2 + \dots + a_{1p}x_p?$$

$$a_1 = \arg \max_a \text{Var}(a^T X)$$

$$= \arg \max_a a^T \Sigma_X a?$$

$$Y_2 = a_2^T X = a_{21}x_1 + a_{22}x_2 + \dots + a_{2p}x_p?$$

$$a_2 = \arg \max_{a, \text{Cov}(Y_1, Y_2) = 0} \text{Var}(a^T X) = \arg \max_{a, a^T \Sigma a = 0} \text{Var}(a^T X)$$

$$\text{Cov}(Y_1, Y_2)$$

$$= \text{Cov}(a_1^T X, a_2^T X)$$

$$= a_1^T \Sigma_X a_2$$

$$Y_3 = a_3^T X = \dots$$

$$a_3 = \arg \max_a \text{Var}(a^T X)$$

$$a_1, \text{Cov}(Y_1, Y_2) = 0$$

$$= \arg \max_a \text{Var}(a^T X)$$

$$a, a^T \Sigma a = 0$$

$$\frac{1}{\sqrt{n}} X^T D^T H^T \cdot \frac{1}{\sqrt{n}} H D X$$

Maximizing the Criteria

$$a_1 = \arg \max_a \text{Var}(a^T X)$$

$$\vec{a}' = c\vec{a} \text{ where } c > 0$$

$\therefore \text{Var}(a^T X)$ could be arbitrary large

The criteria to be maximized is $\max (\mathbf{a}^T \Sigma_X \mathbf{a})$. We can always multiply $Y_1 = \mathbf{a}^T \mathbf{X}$ by a constant $|c| > 1$, which will increase the variance, $\text{var}(cY_1) = \text{var}(\mathbf{c}\mathbf{a}^T \mathbf{X}) = c^2 \text{var}(\mathbf{a}^T \mathbf{X})$. Therefore, we normalize the combination vector $\mathbf{a}^T \mathbf{a} = 1 = L_a^2 = L_a$.

Our problem is to

- find $\mathbf{a}_1$ that maximizes $\mathbf{a}_1^T \Sigma_X \mathbf{a}_1$ subject to $\mathbf{a}_1^T \mathbf{a}_1 = 1$; or equivalently, find $\mathbf{a}_1$ that maximizes variance $\max_{\mathbf{a}_1} \left(\frac{\mathbf{a}_1^T \Sigma_X \mathbf{a}_1}{\mathbf{a}_1^T \mathbf{a}_1} \right) = \text{var}(Y_1)$
- find $\mathbf{a}_2$ that maximizes $\mathbf{a}_2^T \Sigma_X \mathbf{a}_2$ subject to $\mathbf{a}_2^T \Sigma_X \mathbf{a}_1 = 0$ and $\mathbf{a}_2^T \mathbf{a}_2 = 1$; or equivalently $\max_{\mathbf{a}_2^T \Sigma_X \mathbf{a}_1 = 0} \left(\frac{\mathbf{a}_2^T \Sigma_X \mathbf{a}_2}{\mathbf{a}_2^T \mathbf{a}_2} \right) = \text{var}(Y_2)$
- ...
- find $\mathbf{a}_i$ that maximizes $\mathbf{a}_i^T \Sigma_X \mathbf{a}_i$ subject to $\mathbf{a}_i^T \Sigma_X \mathbf{a}_k = 0$ for $k < i$ and $\mathbf{a}_i^T \mathbf{a}_i = 1$;

$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix} \rightarrow 1\text{-dim random vector}$
 Variance Σ_X
 $a_1 = \arg \max_{\|a\|=1} a^T \Sigma_X a$
 $\max_{\|a\|=1} a^T \Sigma_X a = \max_a \frac{a^T \Sigma_X a}{\|a\|^2}$
 $(= \max_{\hat{a} = \frac{a}{\|a\|}} \hat{a}^T \Sigma_X \hat{a})$

The crucial linear algebra result

$$a_2 = \arg \max_{a_2 \perp a_1} \frac{a_2^T B a_2}{a_2^T a_2}$$

Because $a_1 = \frac{u_1}{\|u_1\|}$
 $\frac{a_2^T B a_2}{a_2^T a_2} = \frac{a_2^T \lambda_1 a_1}{a_2^T a_1} = \frac{\lambda_1 a_2^T a_1}{a_2^T a_1} = \lambda_1$

Maximization of Quadratic Forms for Points on the Unit Sphere.

Let $\mathbf{B}$ be a positive definite matrix with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ and associated normalized eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p$. Then

$$\begin{aligned} \max_{\mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}^T \mathbf{B} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} &= \lambda_1 \quad (\text{attained when } \mathbf{x} = \mathbf{u}_1) \\ \min_{\mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}^T \mathbf{B} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} &= \lambda_p \quad (\text{attained when } \mathbf{x} = \mathbf{u}_p) \end{aligned}$$

Moreover,

$$\max_{\mathbf{x} \perp \mathbf{u}_1, \dots, \mathbf{u}_k} \frac{\mathbf{x}^T \mathbf{B} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \lambda_{k+1} \quad (\text{attained when } \mathbf{x} = \mathbf{u}_{k+1}, k = 1, 2, \dots, p-1).$$

$$\arg \max_{a_1, \|a_1\|=1} \text{Var}(a_1^T X) = a_1^T \Sigma_X a_1$$

$$\arg \max_{a_2, \text{Cov}(a_2^T X, a_1^T X) = 0} \text{Var}(a_2^T X) \Rightarrow a_2^T \Sigma_X a_1 = 0$$

$$\arg \max_{a_3, \text{Cov}(a_3^T X, a_1^T X) = 0, \text{Cov}(a_3^T X, a_2^T X) = 0} \text{Var}(a_3^T X)$$

Linear Algebra result:

$$\text{Let } B \geq 0 \text{ with } (\lambda_i, u_i) \text{ for } i=1, \dots, p \text{ s.t. } \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$$

$$\text{then } \max_{x \neq 0} \frac{x^T B x}{x^T x} = \lambda_1, \quad ("=" \text{ when } x = u_1)$$

$$\min_{x \neq 0} \frac{x^T B x}{x^T x} = \lambda_p \quad ("=" \text{ when } x = u_p)$$

$$\max_{x \perp u_1, \dots, u_k} \frac{x^T B x}{x^T x} = \lambda_{k+1} \quad ("=" \text{ holds when } x = u_{k+1})$$

Pf: Let $B = U \Lambda U^T$:

$$\frac{x^T B x}{x^T x} = \frac{x^T U \Lambda U^T x}{x^T U U^T x}$$

$$y = U^T x \Rightarrow \frac{y^T \Lambda y}{y^T y} = \frac{\sum_{i=1}^p \lambda_i y_i^2}{\sum_{i=1}^p y_i^2}$$

$$\max_{x \perp u_1, \dots, u_k} \frac{x^T B x}{x^T x}$$

$$0 = u_i^T x = u_i^T U y = (u_i^T u_1, u_i^T u_2, \dots, u_i^T u_p) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{pmatrix} = (0, 0, \dots, 1, \dots, 0) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{pmatrix} = y_i$$

$$\text{so } x \perp u_1, \dots, u_k \Leftrightarrow y_1 = \dots = y_k = 0$$

$$\arg \max_{a_1, \|a_1\|=1} \text{Var}(a_1^T X) = a_1^T \Sigma_X a_1$$

$$\arg \max_{a_2, \text{Cov}(a_2^T X, a_1^T X) = 0} \text{Var}(a_2^T X) \Rightarrow a_2^T \Sigma_X a_1 = 0$$

$$\arg \max_{a_3, \text{Cov}(a_3^T X, a_1^T X) = 0, \text{Cov}(a_3^T X, a_2^T X) = 0} \text{Var}(a_3^T X)$$

Linear Algebra result:

$$\text{Let } B \geq 0 \text{ with } (\lambda_i, u_i) \text{ for } i=1, \dots, p \text{ s.t. } \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$$

$$\text{then } \max_{x \neq 0} \frac{x^T B x}{x^T x} = \lambda_1, \quad ("=" \text{ when } x = u_1)$$

$$\min_{x \neq 0} \frac{x^T B x}{x^T x} = \lambda_p \quad ("=" \text{ when } x = u_p)$$

$$\max_{x \perp u_1, \dots, u_k} \frac{x^T B x}{x^T x} = \lambda_{k+1} \quad ("=" \text{ holds when } x = u_{k+1})$$

Pf: Let $B = U \Lambda U^T$:

$$\frac{x^T B x}{x^T x} = \frac{x^T U \Lambda U^T x}{x^T U U^T x}$$

$$y = U^T x \Rightarrow \frac{y^T \Lambda y}{y^T y} = \frac{\sum_{i=1}^p \lambda_i y_i^2}{\sum_{i=1}^p y_i^2}$$

$$\max_{x \perp u_1, \dots, u_k} \frac{x^T B x}{x^T x}$$

$$0 = u_i^T x = y_i$$

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_{k+1} \\ y_{k+2} \\ \vdots \\ y_p \end{pmatrix}$$

$$\text{when } x \perp u_1, \dots, u_k \Rightarrow \frac{\sum_{i=k+1}^p \lambda_i y_i^2}{\sum_{i=k+1}^p y_i^2} \leq \lambda_{k+1}$$

$$\text{so } x \perp u_1, \dots, u_k \Leftrightarrow y_1 = \dots = y_k = 0$$

Proof

Proof. Let $\mathbf{U}$ be the orthogonal matrix whose columns are the $(p \times p)$ eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p$ and $\mathbf{\Lambda}$ be the diagonal matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_p$ along the main diagonal. Let $\mathbf{B}^{1/2} = \mathbf{U}\mathbf{\Lambda}^{1/2}\mathbf{U}^\top$ and $\mathbf{y} = \mathbf{U}^\top \mathbf{x}$. Consequently, $\mathbf{x} \neq \mathbf{0}$ implies $\mathbf{y} \neq \mathbf{0}$ $(p \times 1)$

Thus,

$$\begin{aligned} \frac{\mathbf{x}^\top \mathbf{B} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} &= \frac{\mathbf{x}^\top \mathbf{B}^{1/2} \mathbf{B}^{1/2} \mathbf{x}}{\mathbf{x}^\top \underbrace{\mathbf{U} \mathbf{U}^\top}_{(p \times p)} \mathbf{x}} = \frac{\mathbf{x}^\top \mathbf{U} \mathbf{\Lambda}^{1/2} \mathbf{U}^\top \mathbf{U} \mathbf{\Lambda}^{1/2} \mathbf{U}^\top \mathbf{x}}{\mathbf{y}^\top \mathbf{y}} = \frac{\mathbf{y}^\top \mathbf{\Lambda} \mathbf{y}}{\mathbf{y}^\top \mathbf{y}} \\ &= \frac{\sum_{i=1}^p \lambda_i y_i^2}{\sum_{i=1}^p y_i^2} \leq \lambda_1 \frac{\sum_{i=1}^p y_i^2}{\sum_{i=1}^p y_i^2} = \lambda_1 \end{aligned} \quad (1)$$

Proof continued

Setting $\mathbf{x} = \mathbf{u}_1$ gives

$$\mathbf{y} = \mathbf{U}^\top \mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

since

$$\mathbf{u}_k^\top \mathbf{u}_1 = \begin{cases} 1, & k = 1 \\ 0, & k \neq 1 \end{cases}$$

For this choice of $\mathbf{x}$, we have $\mathbf{y}^\top \mathbf{\Lambda} \mathbf{y} / \mathbf{y}^\top \mathbf{y} = \lambda_1 / 1 = \lambda_1$, or

$$\frac{\mathbf{u}_1^\top \mathbf{B} \mathbf{u}_1}{\mathbf{u}_1^\top \mathbf{u}_1} = \mathbf{u}_1^\top \mathbf{B} \mathbf{u}_1 = \lambda_1$$

A similar argument produces the result for finding the minimum of $\mathbf{x}^\top \mathbf{B} \mathbf{x} / \mathbf{x}^\top \mathbf{x}$.

Proof continued

Now, $\mathbf{x} = \mathbf{U}\mathbf{y} = y_1\mathbf{u}_1 + y_2\mathbf{u}_2 + \cdots + y_p\mathbf{u}_p$, so $\mathbf{x} \perp \mathbf{u}_1, \dots, \mathbf{u}_k$ implies

$$0 = \mathbf{u}_i^\top \mathbf{x} = y_1 \mathbf{u}_i^\top \mathbf{u}_1 + y_2 \mathbf{u}_i^\top \mathbf{u}_2 + \cdots + y_p \mathbf{u}_i^\top \mathbf{u}_p = y_i, \quad i \leq k$$

Therefore, for $\mathbf{x}$ perpendicular to the first k eigenvectors $\mathbf{u}_i$, the left-hand side of the inequality in (1) becomes

$$\frac{\mathbf{x}^\top \mathbf{B}\mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{\sum_{i=k+1}^p \lambda_i y_i^2}{\sum_{i=k+1}^p y_i^2}$$

Taking $y_{k+1} = 1, y_{k+2} = \cdots = y_p = 0$ gives the asserted maximum.

Population PC

Result 1

Let Σ be the covariance matrix associated with the vector $\mathbf{X}^\top = (X_1, X_2, \dots, X_p)$. Let Σ have the eigenvector-eigenvalues pairs $(\lambda_1, \mathbf{u}_1), (\lambda_2, \mathbf{u}_2), \dots, (\lambda_p, \mathbf{u}_p)$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$. Then the i^{th} PC is given by

$$Y_i = \mathbf{u}_i^\top \mathbf{X} = u_{i1}X_1 + u_{i2}X_2 + \dots + u_{ip}X_p$$

for $i = 1, 2, \dots, p$. Given this

$$\text{var}(Y_i) = \mathbf{u}_i^\top \Sigma \mathbf{u}_i = \lambda_i$$

and for $i \neq k$

$$\text{cov}(Y_i, Y_k) = \mathbf{u}_i^\top \Sigma \mathbf{u}_k = 0.$$

If some of the λ_i are equal, then the choice of the corresponding coefficient vectors $\mathbf{u}_i$ (and thus Y_i) are not unique.

1st PC:

$$Y_1 = u_1^T X = u_{11}X_1 + \dots + u_{1p}X_p \text{ where } u_1 = \begin{pmatrix} u_{11} \\ u_{12} \\ \vdots \\ u_{1p} \end{pmatrix}$$

$$i\text{-th PC } Y_i = u_i^T X = u_{i1}X_1 + \dots + u_{ip}X_p \text{ where } u_i = \begin{pmatrix} u_{i1} \\ u_{i2} \\ \vdots \\ u_{ip} \end{pmatrix}$$

$$\text{Var}(Y_i) = u_i^T \underbrace{\Sigma_X}_{\lambda_i u_i} u_i = \lambda_i$$

La

$$\begin{aligned} \text{Cov}(Y_i, Y_j) &= u_i^T \Sigma_X u_j \\ &= u_i^T \lambda_j u_j = 0 \end{aligned}$$

$$\begin{array}{ll} Y_1 & \text{Var}(Y_1) = \lambda_1 \\ Y_2 & \text{Var}(Y_2) = \lambda_2 \\ \vdots & \vdots \\ Y_p & \text{Var}(Y_p) = \lambda_p \end{array}$$

$$\text{so } \lambda \perp u_1, \dots, u_k \Leftrightarrow u_1 = \dots = u_k = 0$$

Population PC continued

We can write all of this in terms of matrices:

$$\mathbf{Y} = \mathbf{U}^\top \mathbf{X} \implies \text{cov}(\mathbf{Y}) = \boldsymbol{\Sigma}_Y = \mathbf{U}^\top \boldsymbol{\Sigma}_X \mathbf{U}$$

So,

$$\underbrace{\boldsymbol{\Sigma}_X}_{\text{cov}(\mathbf{X})} = \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^\top \iff \underbrace{\boldsymbol{\Sigma}_Y}_{\text{cov}(\mathbf{Y})} = \boldsymbol{\Lambda} = \mathbf{U}^\top \boldsymbol{\Sigma}_X \mathbf{U} = \text{diag}(\lambda_i)$$

More population PC results

Result 2

Let $\mathbf{X}^\top = (X_1, X_2, \dots, X_p)$ have covariance matrix Σ_X with eigenvalue and eigenvector pairs $(\lambda_1, \mathbf{u}_1), (\lambda_2, \mathbf{u}_2), \dots, (\lambda_p, \mathbf{u}_p)$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$. Let $Y_1 = \mathbf{u}_1^\top \mathbf{X}, Y_2 = \mathbf{u}_2^\top \mathbf{X} \dots Y_p = \mathbf{u}_p^\top \mathbf{X}$ be the PCs. Then

$$\sigma_{11} + \sigma_{22} + \dots + \sigma_{pp} = \sum_{i=1}^p \sigma_{ii} = \lambda_1 + \lambda_2 + \dots + \lambda_p = \sum_{i=1}^p \lambda_i$$

Proof:

$$\sum_{i=1}^p \text{Var}(X_i) = \text{tr}(\Sigma_X) = \text{tr}(\mathbf{U}\Lambda\mathbf{U}^\top) = \text{tr}(\Lambda\mathbf{U}^\top\mathbf{U}) = \text{tr}(\Lambda) = \sum_{i=1}^p \text{Var}(Y_i)$$

1st PC:

PC1 $Y_1 = u_1^T X = u_{11}X_1 + \dots + u_{1p}X_p$ where $u_1 = \begin{pmatrix} u_{11} \\ \vdots \\ u_{1p} \end{pmatrix}$

$$\text{tr} A = \sum_{i=1}^n A_{ii}$$

2nd PC $Y_2 = u_2^T X = u_{21}X_1 + \dots + u_{2p}X_p$ where $u_2 = \begin{pmatrix} u_{21} \\ u_{22} \\ \vdots \\ u_{2p} \end{pmatrix}$

$$\vdots$$

PC3

$\vdots$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_p \end{pmatrix} = \begin{pmatrix} u_1^T X \\ u_2^T X \\ \vdots \\ u_p^T X \end{pmatrix} = U^T X$$

$$\bar{Z} = \begin{pmatrix} 2 & 0.3 & 0.5 & 2 \\ 0.3 & 3 & 1 & 1 \\ 0.5 & 1 & 1 & 1 \\ 2 & 1 & 1 & 4 \end{pmatrix}$$

$$\sum_{i=1}^p \text{Var}(Y_i) = \text{tr}(\bar{Z}_X)$$

$$\text{or } \sum_{i=1}^p \lambda_i$$

1st PC:

PC1 $Y_1 = u_1^T X = u_{11}X_1 + \dots + u_{1p}X_p$ where $u_1 = \begin{pmatrix} u_{11} \\ \vdots \\ u_{1p} \end{pmatrix}$

A symmetric
 $\text{tr} A = \sum_{i=1}^n \lambda_i(A)$

2nd PC $Y_2 = u_2^T X = u_{21}X_1 + \dots + u_{2p}X_p$ where $u_2 = \begin{pmatrix} u_{21} \\ u_{22} \\ \vdots \\ u_{2p} \end{pmatrix}$

$$\vdots$$

PC3

$\vdots$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_p \end{pmatrix} = \begin{pmatrix} u_1^T X \\ u_2^T X \\ \vdots \\ u_p^T X \end{pmatrix} = U^T X$$

$$\text{tr} AB = \text{tr} BA = \text{tr}(U^T U \Lambda)$$

$$= \text{tr}(\Lambda)$$

$$= \sum_{i=1}^p \lambda_i$$

$$\sum_{i=1}^p \text{Var}(Y_i) = \text{tr}(\bar{Z}_X)$$

$$\text{or } \sum_{i=1}^p \lambda_i$$

Proportion of variances accounted for

- The Proportion of total variance due to the k^{th} PC is

$$\frac{\lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_p} = \frac{\lambda_k}{\sum_{i=1}^p \lambda_i} \quad k = 1, \dots, p$$

- We often select q PCs such that the proportions for $k = 1, \dots, q$ sum up as close to 1 (yet not too large of a value for q). The Proportion of Variance accounted for by the first q PCs equals

$$\frac{\sum_{k=1}^q \lambda_k}{\text{trace}(\mathbf{\Sigma}_X)}$$

We try to balance the percent of variance (information) retained and the number of PCs (simplicity). We may want to replace $\mathbf{X}$ by $\mathbf{Y}$.

- Often we're interested interpreting the new variables (i.e., the PCs), so we examine the elements of the $\mathbf{u}_i$'s. The size (magnitude) of the elements of $\mathbf{u}_i$ are an indicator of a variables "importance" to the i^{th} PC...

Correlation between Y_i and X_k

If $Y_1 = \mathbf{u}_1^\top \mathbf{X}$, $Y_2 = \mathbf{u}_2^\top \mathbf{X}$, $\dots$ $Y_p = \mathbf{u}_p^\top \mathbf{X}$ are the PCs obtained from Σ_X , we can use ρ_{Y_i, X_k} to help interpret the contribution of an X_k to Y_i .

$$\begin{aligned}
 \rho_{Y_i, X_k} &= \frac{\text{cov}(Y_i, X_k)}{\sqrt{\lambda_i} \sqrt{\sigma_{kk}}} \\
 &= \frac{\text{cov}(\mathbf{u}_i^\top \mathbf{X}, \mathbf{e}_k^\top \mathbf{X})}{\sqrt{\lambda_i} \sqrt{\sigma_{kk}}} \text{ where } \mathbf{e}_k^\top = (0, \dots, \underbrace{1}_{k^{\text{th}} \text{ position}}, 0, \dots, 0) \\
 &= \frac{\mathbf{e}_k^\top \Sigma \mathbf{u}_i}{\sqrt{\lambda_i} \sqrt{\sigma_{kk}}} \\
 &= \frac{\mathbf{e}_k^\top (\lambda_i \mathbf{u}_i)}{\sqrt{\lambda_i} \sqrt{\sigma_{kk}}} \\
 &= \frac{u_{ik} \sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}
 \end{aligned}$$

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Sample Principal Components

The Algebra is the same as in population principal components.

- $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are n independent observations from a population with μ and Σ .
- $\bar{\mathbf{x}}_{p \times 1}$ = sample mean vector.
- Sample covariance matrix.

$$\begin{aligned} \mathbf{S}_{p \times p} &= \frac{1}{n-1} \sum_i (\mathbf{x}_i - \bar{\mathbf{x}}) (\mathbf{x}_i - \bar{\mathbf{x}})^T = \frac{1}{n-1} \left(\sum_i \mathbf{x}_i \mathbf{x}_i^T - n \bar{\mathbf{x}} \bar{\mathbf{x}}^T \right) \\ &= \frac{1}{n-1} \mathbf{X}^T \left(\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) \mathbf{X}. \end{aligned}$$

- $\mathbf{S}$ has eigenvalue/vector pairs $(\hat{\lambda}_1, \hat{\mathbf{u}}_1), \dots, (\hat{\lambda}_p, \hat{\mathbf{u}}_p)$ where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_p$
- The $\hat{}$ indicates these are estimates of population values.

Algebra of Sample PC continued

- The i^{th} sample principal component is given by

$$\hat{\mathbf{y}}_i = \hat{\mathbf{u}}_i^\top \mathbf{x} = \hat{u}_{i1}x_1 + \hat{u}_{i2}x_2 + \cdots + \hat{u}_{ip}x_p$$

- The i^{th} PC sample variance = $\text{var}(\hat{y}_i) = \hat{\lambda}_i$ for $i = 1, \dots, p$.
- The PC sample covariances = $\text{cov}(\hat{y}_i, \hat{y}_k) = 0$ for all $i \neq k$.
- Total sample variance

$$\text{trace}(\mathbf{S}) = \text{tr}(\mathbf{S}) = \sum_{i=1}^p s_{ii} = \sum_{i=1}^p \hat{\lambda}_i$$

- Proportion of total sample variance accounted for by the i^{th} PC

$$\frac{\hat{\lambda}_i}{\sum_{k=1}^p \hat{\lambda}_k}$$

- Correlations between $\hat{y}_i$ and x_k is $r_{\hat{y}_i, x_k} = \frac{\sqrt{\hat{\lambda}_i}}{\sqrt{s_{kk}}} \hat{u}_{ik}$

Example

These data are the national record times for men before the 1984 Olympics. The record times for eight races (i.e., $p = 8$) are listed for 55 countries (i.e., $n = 55$). The times are recorded for the following races:

- 100 m: Record time for 100 m race in seconds
- 200 m : Record time for 200 m race in seconds
- 400 m : Record time for 400 m race in seconds
- 800 m : Record time for 800 m race in minutes
- 1500 m : Record time for 1500 m race in minutes
- 5 K : Record time for 5000 m race in minutes
- 10 K : Record time for 10000 m race in minutes
- Marathon: Record time for the Marathon (approx. 26 miles) in

Summary statistics

Summary Statistics for each variable are given below:

	100 m	200 m	400 m	800 m	1500 m	5 K	10 K	Marathon
$\bar{x}$	10.47	20.90	46.44	1.79	3.70	13.85	28.99	136.62
sd	0.35	0.64	1.46	0.06	0.16	0.80	1.81	9.23

Covariance Matrix

	m100	m200	m400	m800	m1500	K5	K10	Mara.
m100	0.12	0.20	0.43	0.01	0.03	0.17	0.40	1.68
m200	0.20	0.41	0.79	0.03	0.07	0.35	0.81	3.54
m400	0.43	0.79	2.12	0.08	0.18	0.90	2.07	9.47
m800	0.01	0.03	0.08	0.004	0.00	0.04	0.10	0.47
m1500	0.03	0.07	0.18	0.01	0.02	0.11	0.26	1.24
K5	0.17	0.35	0.90	0.04	0.11	0.64	1.41	6.89
K10	0.40	0.81	2.07	0.10	0.26	1.41	3.26	15.73
Marathon	1.68	3.54	9.47	0.47	1.24	6.89	15.73	85.13

Eigenvalues of Σ

Total Variance = 91.738234815

Eigenvalues of the Covariance Matrix:

	Eigenvalue	Difference	Proportion	Cumulative
1	89.914	88.500	0.980	0.9801
2	1.412	1.153	0.015	0.9955
3	0.260	0.150	0.003	0.9983
4	0.109	0.082	0.001	0.9995
5	0.027	0.015	0.000	0.9998
6	0.013	0.010	0.000	1.0000
7	0.002	0.002	0.000	1.0000
8	0.000		0.000	1.0000

First three eigenvectors of Σ

Race	PC loadings		
	Prin1	Prin2	Prin3
m100	0.02	0.21	-.03
m200	0.04	0.36	-.02
m400	0.11	0.83	-.38
m800	0.01	0.02	0.01
m1500	0.01	0.04	0.05
K5	0.08	0.13	0.34
K10	0.18	0.30	0.85
Marathon	0.97	-.18	-.14

The 1st principal component is essentially the marathon, because it has by far the largest variance 85.13 compared to the next largest which is 3.26 (the 10K)

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When Variances are very different

Principal Components obtained from Standardized Variables:
If we use standardized variables (“z-scores”)

$$Z_1 = \frac{X_1 - \mu_1}{\sqrt{\sigma_{11}}}$$

$$Z_2 = \frac{X_2 - \mu_2}{\sqrt{\sigma_{22}}}$$

$$\vdots$$

$$Z_p = \frac{X_p - \mu_p}{\sqrt{\sigma_{pp}}}$$

or in matrix notation

$$\mathbf{Z} = \underbrace{\mathbf{D}_\sigma^{-1/2}}_{\text{diag}(1/\sqrt{\sigma_{ii}})} \underbrace{(\mathbf{X} - \boldsymbol{\mu})}_{p \times 1} = \mathbf{D}_\sigma^{-1/2} \mathbf{X} - \mathbf{D}_\sigma^{-1/2} \boldsymbol{\mu}$$

PCs of standardized variables

We know that

$$E(\mathbf{Z}) = E\left(\mathbf{D}_\sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})\right) = \mathbf{D}_\sigma^{-1/2} \underbrace{E(\mathbf{X})}_{\boldsymbol{\mu}} - \mathbf{D}_\sigma^{-1/2}\boldsymbol{\mu} = \mathbf{0}$$

and

$$\boldsymbol{\Sigma}_Z = \mathbf{D}_\sigma^{-1/2} \boldsymbol{\Sigma}_X \mathbf{D}_\sigma^{-1/2} = \mathcal{R}$$

PCs of standardized variables

The i^{th} PC of the standardized variables $\mathbf{Z}^\top = (Z_1, Z_2, \dots, Z_p)$ with $\boldsymbol{\Sigma}_Z = \mathcal{R}$ is given by

$$\tilde{Y}_i = \tilde{\mathbf{u}}_i^\top \mathbf{Z} = \tilde{\mathbf{u}}_i^\top \left(\mathbf{D}_\sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu}) \right) \quad \text{for } i = 1, 2, \dots, p$$

where $\tilde{\mathbf{u}}_i$ is the i^{th} eigenvector and $\tilde{\lambda}_i$ is the i^{th} eigenvalue of $\mathcal{R}$.

PCs of standardized variables

- We have

$$\sum_{i=1}^p \text{var}(Z_i) = \sum_{i=1}^p \text{var}(\tilde{Y}_i) = \text{trace}(\mathcal{R}) = p$$

and

$$\rho_{\tilde{Y}_i, Z_k} = \sqrt{\tilde{\lambda}_i} \tilde{u}_{ik}$$

- Almost always

$$\lambda_i \neq \tilde{\lambda}_i \quad \text{and} \quad \mathbf{u}_i \neq \tilde{\mathbf{u}}_i;$$

The PCs from Σ_X are not the same as PCs from $\mathcal{R}$. We take a look again at the previous example situation where standardization makes a difference. This will be the case when the scales of the X variables are (substantially or vastly) different and they are not comparable.

The correlation matrix

	m100	m200	m400	m800	m1500	K5	K10	Mara.
m100	1.00	.92	.84	.75	.70	.61	.63	.51
m200	.92	1.00	.85	.80	.77	.69	.69	.59
m400	.84	.85	1.00	.87	.83	.77	.78	.70
m800	.75	.80	.87	1.00	.91	.86	.86	.80
m1500	.70	.77	.83	.91	1.00	.92	.93	.86
K5	.61	.69	.77	.86	.92	1.00	.97	.93
K10	.63	.69	.78	.86	.93	.97	1.00	.94
Marathon	.51	.59	.70	.80	.86	.93	.94	1.00

Eigenvalues of the correlation matrix

Eigenvalues of the Correlation matrix

	Eigenvalue	Difference	Proportion	Cumulative
1	6.62	5.745	0.828	0.828
2	0.87	0.718	0.110	0.938
3	0.15	0.035	0.020	0.957
4	0.12	0.044	0.025	0.973
5	0.08	0.012	0.010	0.983
6	0.06	0.022	0.018	0.991
7	0.04	0.024	0.015	0.997
8	0.02		0.002	1.000

Total variance = 8. The first 2 principal components account for 93.8% of the total variance.

The Eigenvectors of the correlation matrix

The First Two Eigenvectors

Race	PC Loadings	
	1	2
100 m	.318	.567
200 m	.337	.462
400 m	.356	.248
800 m	.369	.012
1500 m	.373	-.140
5 K	.364	-.312
10k	.367	-.307
Marathon	.342	-.439

Graph of countries component scores

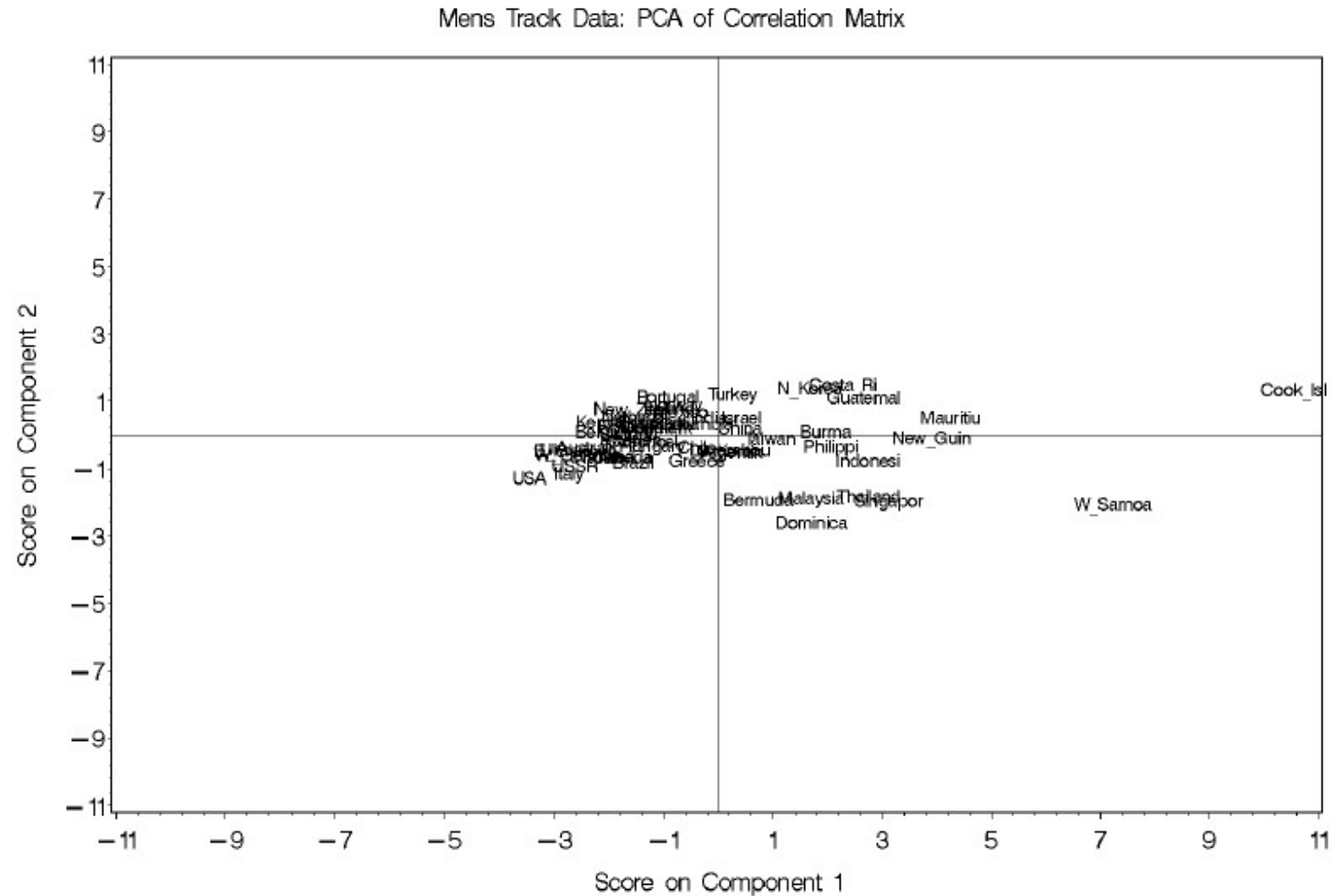
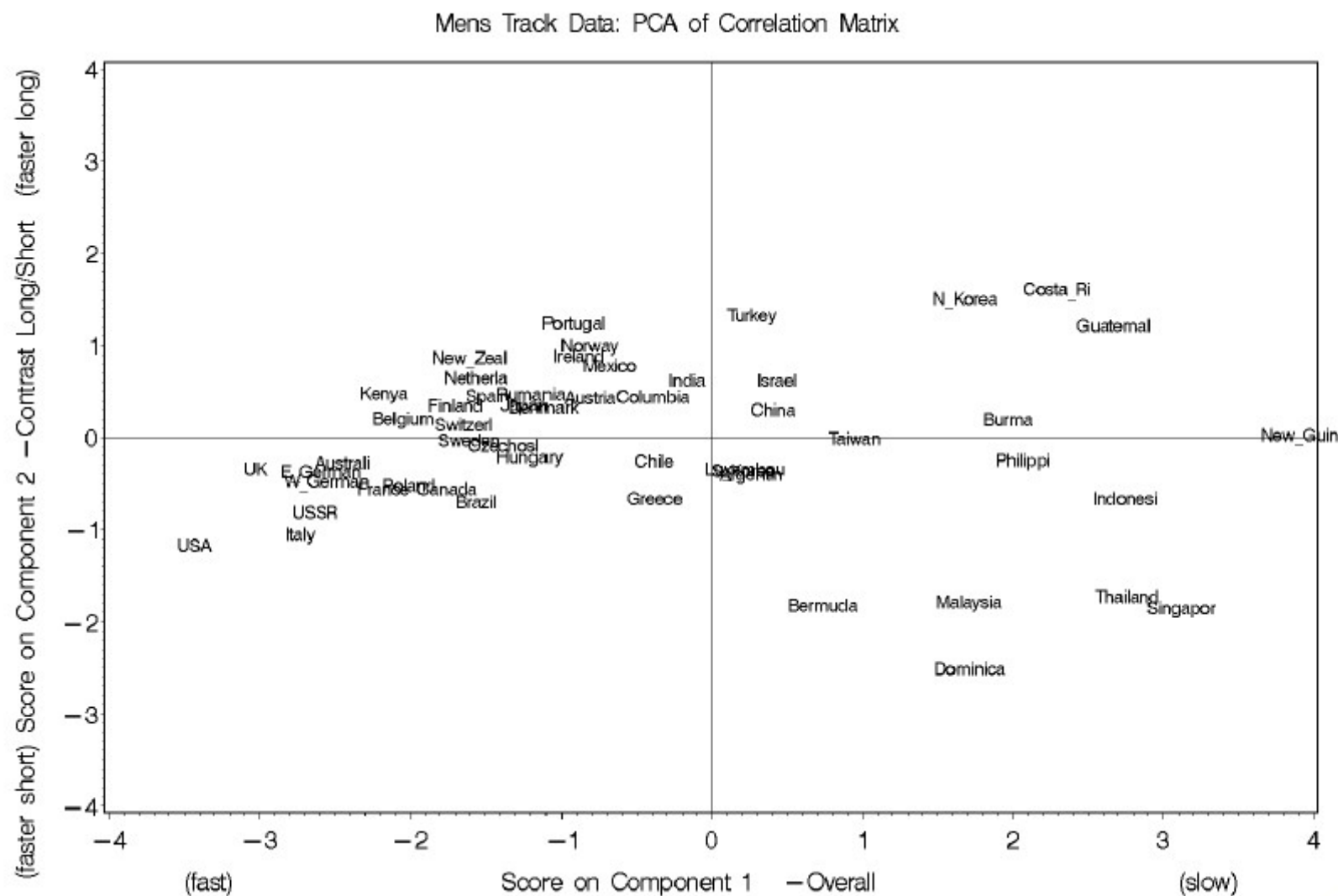


Figure: PC scores for the first two components

Graph of countries component scores (zoom in)



Interpretation

Interpretation?

- First component: An overall measure - High values on this component indicate slower runners.
- Second component: Contrast long and short races:
 - Large values indicate slower on short races than long ones.
 - Small values indicate faster on short races than long ones.
 - Value near zero means that tend to be similar on short and long races (could be slow, fast, or somewhere in between on all races).

Correlations(Variable & Components)

The correlations between the standardized variables and values on the principal components equal

$$r_{Z_k, Y_i} = \sqrt{\lambda_i} u_{ik} \quad (\text{e.g., } \sqrt{6.622}(.318) = .82)$$

	Components	
Race	1	2
100 m	.82	.53
200 m	.87	.43
400 m	.92	.23
800 m	.95	.01
1500 m	.96	-.13
5 K	.94	-.29
10k	.94	-.29
Marathon	.88	-.41

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Sampling Theory

If $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ is a sample from $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then the sample principal components

$$\hat{Y}_i = \hat{\mathbf{u}}_i^\top (\mathbf{X} - \bar{\mathbf{X}})$$

are observations ("realizations") of the population principal components

$$Y_i = \mathbf{u}_i^\top (\mathbf{X} - \boldsymbol{\mu}) \quad \begin{matrix} V = (\mathbf{u}_1, \dots, \mathbf{u}_k) \\ Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_k \end{pmatrix} = V^\top (\mathbf{X} - \boldsymbol{\mu}) \end{matrix}$$

and since $\hat{y}_i$ is a linear combination of $\mathbf{x}$ which come from $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\hat{\mathbf{Y}} = \begin{pmatrix} \hat{Y}_1 \\ \hat{Y}_2 \\ \vdots \\ \hat{Y}_p \end{pmatrix} \sim \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Lambda})$$

$$\begin{matrix} S \rightarrow \Sigma \\ \hat{\lambda}_i \rightarrow \lambda_i \\ \hat{\mathbf{u}}_i \rightarrow \mathbf{u}_i \end{matrix}$$

$$\text{then } V^\top \Sigma V = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_k \end{pmatrix} = \boldsymbol{\Lambda}$$

$$\text{so } Y \sim \mathcal{N}(0, \boldsymbol{\Lambda})$$

where $\boldsymbol{\Lambda} = \text{diag}(\lambda_i)$.

Sampling Theory continued

Assume $\mathbf{X}_j \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ i.i.d. for $j = 1, 2, \dots, n$. $\boldsymbol{\Sigma}$, which is unknown, has eigenvalues $\lambda_1 > \lambda_2 > \dots > \lambda_p$ (assumption) with associated eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p$. For n "very large" n

- $\hat{\lambda}_i$ is independent of its corresponding $\hat{\mathbf{u}}_i$
- $\sqrt{n}(\hat{\boldsymbol{\lambda}} - \boldsymbol{\lambda}) \approx \mathcal{N}_p(\mathbf{0}, 2\boldsymbol{\Lambda}^2)$ or that

$$\hat{\boldsymbol{\lambda}} \approx \mathcal{N}_p\left(\boldsymbol{\lambda}, \frac{2}{n}\boldsymbol{\Lambda}^2\right)$$

where $\hat{\boldsymbol{\lambda}}$ are eigenvalues of $\mathbf{S}$, and $\boldsymbol{\lambda}$ are eigenvalues of $\boldsymbol{\Sigma}$. So

$$\hat{\lambda}_i \approx \mathcal{N}_1\left(\lambda_i, \frac{2}{n}\lambda_i^2\right) \quad \text{for } i = 1, \dots, p$$

$X_i \in \mathbb{R}^p$ i.i.d. $EX_i = \mu$, $Cov(X_i) = \Sigma$

$$\hat{\lambda} = \begin{pmatrix} \hat{\lambda}_1 \\ \hat{\lambda}_2 \\ \vdots \\ \hat{\lambda}_p \end{pmatrix} \quad \lambda = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_p \end{pmatrix}$$

$$\hat{\lambda}_i \quad \hat{\mu}_i$$

$$\lambda_i \quad \mu_i$$

$$X \quad h(x)$$

$$\sqrt{n}(\hat{\lambda} - \lambda) \approx N_p(0, 2\Lambda^2) \text{ where } \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_p \end{pmatrix}$$

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^T$$

$$\hat{\lambda}_1 \dots \hat{\lambda}_p$$

$\hat{\lambda}_i$ is h(x) of

X where $X \sim \text{Wishart}$
reason that: $\sqrt{n}(\hat{\lambda} - \lambda) \approx N_p(0, 2\Lambda^2)$

$$\sqrt{n}(\bar{X} - \mu) \rightarrow N(0, \Sigma)$$



Sampling Theory continued

$$\hat{\mathbf{u}}_i^T \hat{\mathbf{u}}_i = 1 \quad \text{for } \forall i$$

And $\sqrt{n}(\hat{\mathbf{u}}_i - \mathbf{u}_i) \approx \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma}_i)$ where

$$\boldsymbol{\Sigma}_i = \lambda_i \sum_{k \neq i} \frac{\lambda_k}{(\lambda_k - \lambda_i)^2} \mathbf{u}_k \mathbf{u}_k^T$$

Note: $\boldsymbol{\Sigma}_i$ is not diagonal, and the Eigenvectors are not independent.

Inference on λ_i

Since the $\hat{\lambda}_i^\top$'s are asymptotically (very large n) independent and normal with mean λ_i and variance $(2/n)\lambda_i^2$, a $(1 - \alpha)100\%$ confidence interval for λ_i is

$$\frac{\hat{\lambda}_i}{\left(1 + z_{\alpha/2} \sqrt{2/n}\right)} \leq \lambda_i \leq \frac{\hat{\lambda}_i}{\left(1 - z_{\alpha/2} \sqrt{2/n}\right)}$$

Handwritten notes:

$$\frac{\hat{\lambda}_i - \lambda_i}{\sqrt{\frac{2}{n}} \lambda_i} \approx \mathcal{N}(0,1) \quad P\left(z_{\frac{\alpha}{2}} \leq \frac{\hat{\lambda}_i}{\sqrt{\frac{2}{n}} \lambda_i} - \frac{1}{\sqrt{\frac{2}{n}}} \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left[-\sqrt{\frac{2}{n}} z_{\frac{\alpha}{2}} \leq \frac{\hat{\lambda}_i}{\lambda_i} - 1 \leq \sqrt{\frac{2}{n}} z_{\frac{\alpha}{2}}\right] = 1 - \alpha$$

where $z_{\alpha/2}$ is the upper $(\alpha/2)$ -percentile of $\mathcal{N}(0, 1)$.

If we want to draw inference on several λ_i simultaneously, use a Bonferroni-type procedure and use $z_{\alpha/(2m)}$ where m = number of intervals you plan to construct.

Handwritten note:

$$P\left[-\sqrt{\frac{2}{n}} z_{\frac{\alpha}{2}} \leq \frac{\hat{\lambda}_i}{\lambda_i} - 1 \leq \sqrt{\frac{2}{n}} z_{\frac{\alpha}{2}}\right] = 1 - \alpha$$

For any $x \in \mathbb{R}^p$ $\dim(\mathcal{P}_U x) \leq q$

$$U^T x_i$$

$$P_{\text{Col}(A)} = A(A^T A)^{-1} A^T$$

Another interpretation:

$$\hat{U} = \arg \min_{U \in \mathbb{R}^{p \times q}, U^T U = I_q} \sum_{i=1}^n \|x_i - P_U x_i\|_2^2$$

$$P_{\text{Col}(A)} y$$

$$= A(A^T A)^{-1} A^T y \in \text{Col}(A)$$

$$\hat{U} = (\hat{u}_1, \dots, \hat{u}_q) \text{ where } \hat{u}_i \text{ are}$$

$$= (a_1, \dots, a_q) \begin{pmatrix} c_1 \\ \vdots \\ c_q \end{pmatrix}$$

$$= c_1 a_1 + \dots + c_q a_q$$

S

$$= \underbrace{U U^T}_{I} x = (u_1, \dots, u_q) \begin{pmatrix} u_1^T x \\ \vdots \\ u_q^T x \end{pmatrix}$$

$$\Phi = SA$$

$$((SA)^T (SA))^{-1} A^T S^T y = (A^T S^T S A)^{-1} A^T S^T y$$

$$(x^T x)^{-1} x^T y$$

PCA: new variables

① what is PCA?

② How to maximize Var. (Quadratic form) $a^T Z a$

③.